SOFTWARE ENGINEERING PROJECT

ZenStudy

SMART STUDY LIBRARY MANAGEMENT SYSTEM

Team-051 · DeerInTheCode

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A Real-World Study Library Environment

- Study libraries provide dedicated, quiet spaces for students to learn and prepare.
- Core operations include memberships, seat allocation, bookings, attendance and payments.
- Many workflows rely on manual registers, spreadsheets and informal communication.
- Students, library staff and administrators have different operational needs.



The project documentation identifies the domain as a Study Library Management System, focused specifically on memberships, bookings, attendance, payments and daily operations.

PROBLEMS FACED

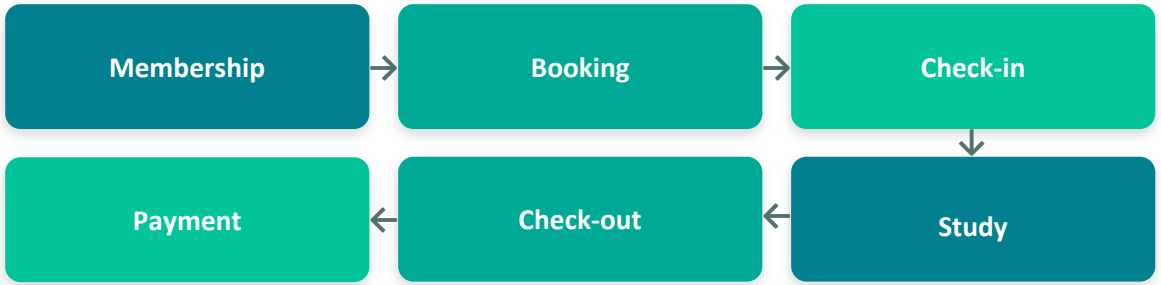
Key Pain Points Discovered

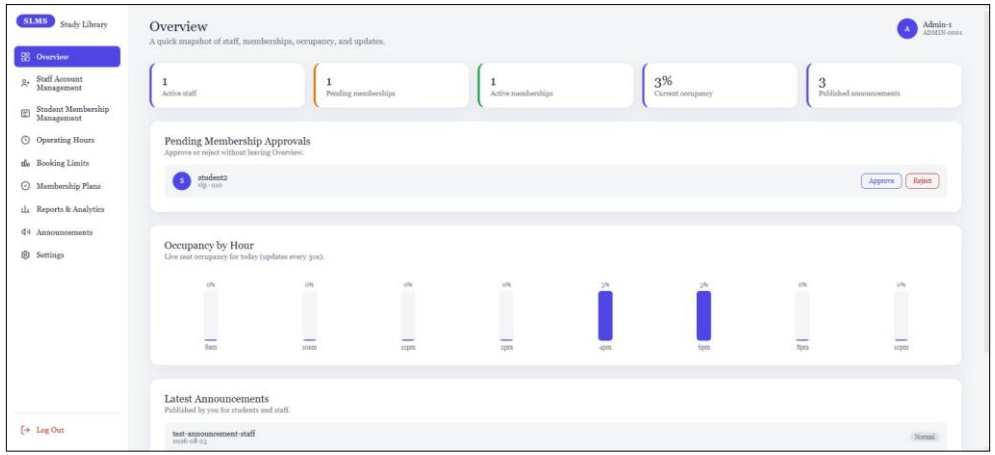
These pain points were identified through the project's user research and stakeholder analysis.

- ✕ No real-time visibility of available seats and study spaces
- ✕ Manual membership verification and seat allocation
- ✕ Difficult management of bookings, cancellations and extensions
- ✕ Repetitive attendance and check-in/check-out processes
- ✕ Separate and manual payment records
- ✕ Difficulty identifying no-shows and releasing unused seats
- ✕ No centralized system for complaints and communication
- ✕ Limited visibility into occupancy, memberships and revenue

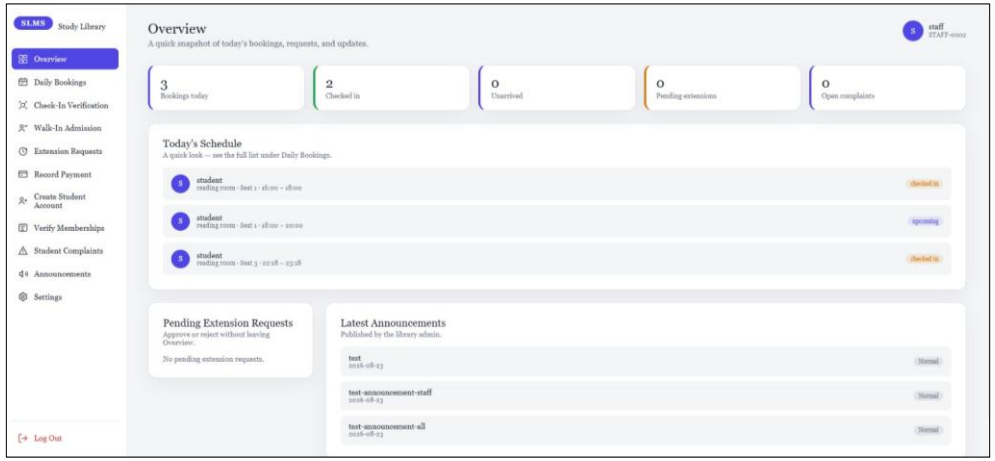
ZenStudy –Study-Library Management

- Complete digital platform for study-library operations
- Real-time seat availability and centralized booking management
- Role-based experience for Students, Library Staff and Admins
- Centralized attendance, payments, complaints and communication
- Administrative reports, analytics and configuration

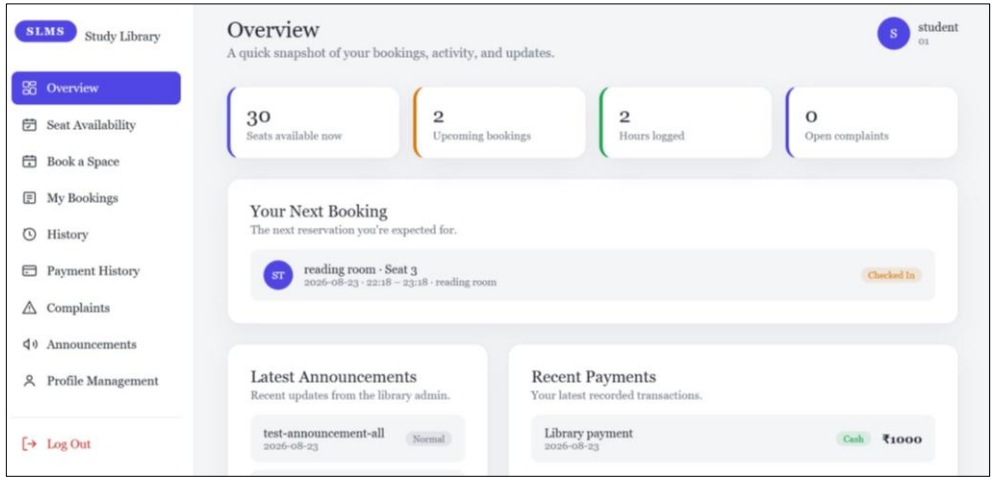




Admin dashboard



Staff dashboard



Student dashboard

THREE USER ROLES

Who Uses ZenStudy?

The system explicitly defines these three roles and provides dedicated role-based dashboards.

STUDENT

- Membership
- Seat availability
- Bookings
- Attendance
- Payments

LIBRARY STAFF

- Student registration
- Bookings
- Check-in
- Seat allocation
- Payments

ADMIN

- Staff
- Memberships
- Library settings
- Announcements
- Analytics

KEY FEATURES & USP

What Makes ZenStudy Special?

These capabilities are directly reflected in the system component design.

01

Real-Time Seat Management

Check availability and prevent double-booking

02

Smart Booking Lifecycle

Book, modify, cancel and extend study sessions

03

Digital Check-In & Attendance

Unique check-in codes with real-time occupancy

04

No-Show Management

Release unused seats for reallocation and rebooking

05

Centralized Membership & Payments

Manage plans, memberships and transaction history

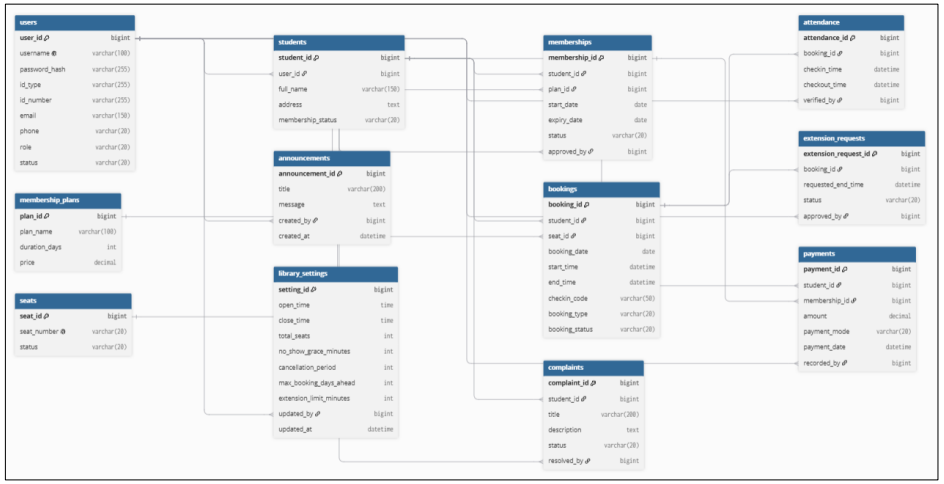
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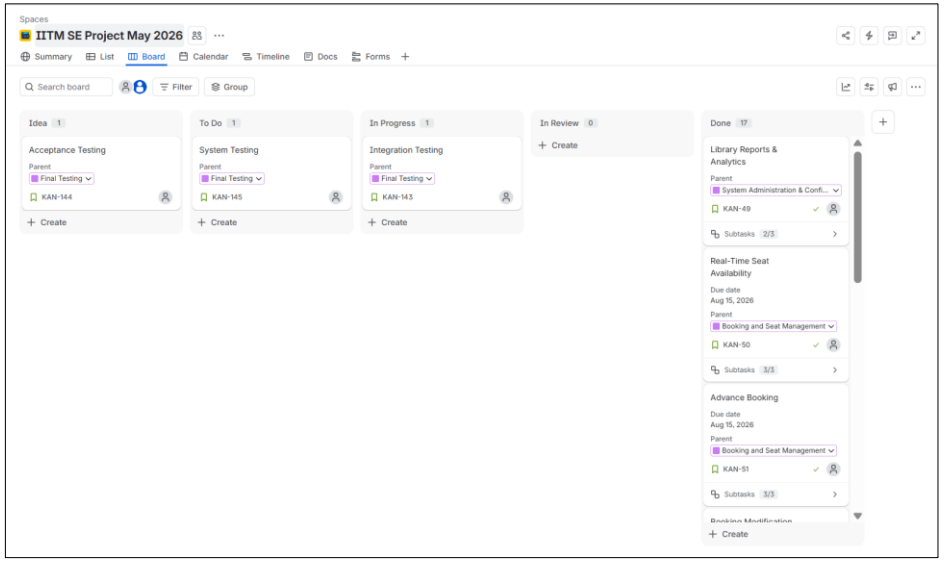
Unified Communication

Complaints, announcements and notifications in one place

System Design & Technology

- Role-based architecture with dedicated Student, Staff and Admin experiences
- UML Class Diagram & ER Diagram defining the system and database structure
- API-driven backend covering authentication, bookings, attendance, payments and administration
- Relational data model connecting users, memberships, bookings, seats, attendance and payments
- Automated API testing with Pytest covering authentication, authorization and business workflows
- Agile execution with Jira/GitHub workflow, Scrum and Kanban





Thank You!

ZenStudy

Making Every Study Session Count

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